

AIGR Deep Research — K9IFT7 (DESRO): beta-defensin orthology and CCR6 binding

Gene: K9IFT7 / *Desmodus rotundus* (vampire bat, NCBITaxon:9430) — GenBank JAA44743.1, RefSeq XP_071076263.1 **Focus:** function_assignment — does K9IFT7 directly have GO:0031731 "CCR6 chemokine receptor binding"? **Source:** genes/DESRO/K9IFT7/K9IFT7-ai-review.yaml (free-text seed)

Executive Judgment

Verdict: PARTIALLY SUPPORTED (homology-plausible but unverified; not the core function).

- **Orthology to the beta-defensin-1 (DEFB1) subfamily is well supported.** K9IFT7 is a canonical 67-aa beta-defensin (Pfam PF00711, InterPro IPR001855, PANTHER PTHR21388:SF9 "BETA-DEFENSIN 1", SUPFAM SSF57392), with the intact 6-cysteine motif (inter-Cys gaps 7-5-10-7-1), cationic net charge (+4), and highest mature-peptide identity to human **DEFB1 (hBD-1, 59.1%)** among the human panel tested (hBD-3 53.7%, DEFB118 52.3%, hBD-2 40.9%, hBD-4 36.4%). Identity is only *moderate*, and bat beta-defensins are an expanded, fast-evolving multigene family, so strict 1:1 orthology to DEFB1 is not proven.

- **CCR6 binding (GO:0031731) is a phylogenetic prediction, evidence code IEA:TreeGrafter** — grafted onto the DEFB1 subfamily. There is **no direct assay of K9IFT7 or any bat defensin.**

- The prediction is **plausible** because the closest human ortholog, DEFB1, has *direct* experimental CCR6 interaction

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and K9IFT7 retains a documented CCR6-chemotaxis determinant (N-terminal Leu;

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- **But it is contested and non-core.** Whether CCR6 is a genuine functional beta-defensin receptor has been directly challenged

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beta-defensins also use CCR2

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and receptor usage varies by defensin/cell type (hBD-1 ≠ hBD-2 for neutrophils,

P 15009427).

The gene's *documented* role (Vampirome,

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is **salivary-gland antimicrobial defense**, not chemotaxis.

Bottom line for the curator: the CCR6-binding term is defensible only as a low-confidence, homology-transferred (IEA), **non-core** molecular function. Do **not** upgrade it to an experimental code; do **not** treat the absence of a bat assay as refutation. The best-supported annotations are antimicrobial defense response and extracellular localization.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Claim tested	Key finding	Context	Confidence/ limitations
UniProt K9IFT7 (DB record)	computational/ database	Qualifies	Term provenance	GO:0031731 is IEA:TreeGraft (phylogenetic graft to PANTHER SF9 "beta- defensin 1"); no experimental code	<i>D. rotundus</i> TrEMBL entry	High tier IEA; w GO ev tier
This report (NW alignment)	structural/ evolutionary (computational)	Supports (orthology)	Which human defensin is closest	Mature K9IFT7 most identical to DEFB1/hBD-1 (59.1%) ; canonical 6-Cys motif, net +4, N- term Leu	in-silico vs human panel	Model: simple alignm not a phylo tree recon
 23411029 (Vampirome)	localization/ expression (primary)	Qualifies/ Competing	Documented function	Bat salivary accessory submandibular gland antimicrobial (with lysozyme, lactotransferrin); no CCR6 assay	<i>D. rotundus</i> glands, RNA- seq + LC-MS/ MS	High f antim secret silent CCR6
 10521347 (Yang 1999, Science)	direct assay/ binding (primary)	Supports (family- level)	Beta- defensin- CCR6 binding	Human beta- defensin chemotactic via CCR6; competitively displaced ¹²⁵ I- LARC/CCL20 on CCR6 transfectants	CCR6- transfected cells, iDC, memory T cells	High; used v prima hBD-2 DEFB

Citation	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Claim tested	Key finding	Context	Confidence/ limitations
				(foundational MF evidence)		
P 25122636 (Diao 2014)	direct interaction (primary)	Supports	Ortholog CCR6 binding	Human DEFB1 interacts with CCR6 and triggers Ca ²⁺ mobilization	Human sperm	High for DEFB1; different tissue than H
PubMed (no hits)	absence of evidence	Qualifies	Bat defensin function	No functional characterization of any chiropteran/bat beta-defensin (CCR6 or antimicrobial assay) was found	Chiroptera	Confirmed; annot rests e on cro mamr homol
P 29207656	direct assay (primary)	Supports	CCR6 as DEFB1-linked receptor	CCR6 required for ligand-induced CatSper activation	Human sperm	Corroborates DEFB1 axis
P 20022113 (Tyrrell 2010)	mutant/structure-function (primary)	Supports (plausibility)	CCR6 determinants	N-terminal Leu/Ile essential for CCR6 chemotaxis; K9IFT7 mature N-term = Leu	Defb14 peptide derivatives	Determinants present; K9IFT7 plausibility not pr
P 17705135 (Soruri 2007)	direct assay (primary)	Competing/Refutes (family-level)	Is CCR6 a real beta-defensin receptor	"CCR6 is not a functional receptor for beta-defensins"; CCR6-transfectants unresponsive to hBD-2/-3	Human/murine cells, CCR6 transfectants	Directly conflicting; challenge the w/ annot basis

Citation	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Claim tested	Key finding	Context	Confidence/ limitations
P 20483750 (Röhrli 2010)	direct interaction (primary)	Qualifies/ Competing	Receptor exclusivity	hBD-2/-3 (and mouse orthologs) bind CCR2 , not only CCR6	HEK293- CCR2, human monocytes	Shows multiple receptor CCR6 obligatory
P 15009427 (Niyonsaba 2004)	direct assay (primary)	Qualifies	hBD-1 vs hBD-2 receptor usage	"hBD-2, but not hBD-1" chemoattracts neutrophils via CCR6	TNF-treated human neutrophils	Receptor activation defense cell-type specific
P 21434867 (Morgera 2011)	cellular assay (primary)	Qualifies	CCR6 dependence	iDC responses to hBD2 occur via "CCR6- dependent and -independent" mechanisms	Human iDC/ monocytes	CCR6 contributes

GO Curation Implications

- **GO:0031731 "CCR6 chemokine receptor binding" (MF):** Evidence supports retaining it **only as a low-confidence, homology-based (IEA) prediction flagged non-core**. It should **not** be generalized to bare "protein binding" (uninformative) nor promoted to an experimental code. A curator may reasonably choose to (a) keep the IEA as-is with a not-core note, or (b) down-weight/remove it given the direct family-level conflict

([P 17705135](#))
and the lack of any bat/DEFB1-ortholog-specific salivary assay. **Lead, requires curator verification.**

- **Better-supported annotations to prioritize:** GO:0050830 / GO:0050829 (defense response to Gram-positive/negative bacterium, BP), GO:0002227 (innate immune response in mucosa,

BP), GO:0005576 (extracellular region, CC) — all consistent with the primary Vampirome evidence

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and the Pfam/InterPro defensin identity. These represent the gene product's core function.

- If any experimental-code MF for receptor binding is ever added, it must derive from an assay of K9IFT7 (or a demonstrated bat DEFB1 ortholog), not transfer from human DEFB1.

Mechanistic Scope

The tested molecular function is **direct binding of the secreted mature beta-defensin peptide to the CC chemokine receptor CCR6** (a GPCR), which for human beta-defensins couples to Gai and drives chemotaxis of immature dendritic cells / memory T cells and Ca²⁺ mobilization. This is distinct from, and must be separated from: - **Antimicrobial activity** (membrane-disruptive killing of bacteria/fungi) — the evidenced core role in bat saliva. - **Salivary/glandular expression** (localization, not activity). - **Chemotaxis via other receptors** (CCR2, or CCR6-independent membrane effects). For K9IFT7 specifically, CCR6 binding is *inferred by homology*, not demonstrated; the antimicrobial role is *documented*.

Conflicts and Alternatives

1. Family-level receptor conflict:

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 directly reports CCR6 is *not* a functional beta-defensin receptor — a fundamental challenge to the annotation's premise.

2. Multiple/alternative receptors: CCR2 usage

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 and CCR6-independent mechanisms

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P 21434867)
 mean "CCR6 binding" may overstate specificity.

3. Defensin-specific variability: hBD-1 behaves differently from hBD-2

([P 15009427](#));
transferring a CCR6 phenotype across the whole SF9 subfamily risks paralog over-annotation.

4. Orthology uncertainty: 59% mature identity to DEFB1 with an expanded bat beta-defensin repertoire means K9IFT7 may be a bat-specific paralog rather than a true DEFB1 ortholog; TreeGraft subfamily assignment can misplace fast-evolving lineage-specific defensins.

5. Context mismatch: the strongest DEFB1–CCR6 evidence is in human sperm

([P 25122636](#));
29207656), a context unrelated to a bat salivary antimicrobial.

Knowledge Gaps

Gap	What was checked	Why it matters	Resolving evidence
No assay of K9IFT7 (or any bat defensin) vs CCR6	PubMed; Vampirome	Annotation rests entirely on homology transfer	Direct binding/chemotaxis assay of recombinant K9IFT7 on CCR6 ⁺ cells
True orthology (DEFB1 vs bat-specific paralog)	NW identity, PANTHER SF9	Determines validity of subfamily transfer	Phylogenetic tree reconciliation of chiropteran beta-defensin cluster; synteny
Correct mature N-terminus / cleavage	UniProt Chain 24-67 vs domain 34-66	N-terminal Leu is a CCR6 determinant; propiece removal could alter it	Proteomic N-terminal sequencing of the salivary peptide
Whether CCR6 is a genuine receptor at all	P 17705135 vs 25122636	Field disagreement undermines the term	Reproducible receptor-transfectant binding across labs

Discriminating Tests

1. **Direct receptor assay:** recombinant/synthetic K9IFT7 (correctly folded, 3 disulfides) in a CCR6-transfectant binding + chemotaxis/ Ca^{2+} -flux assay, with CCR6-knockout and CCL20 competition controls.
 2. **Receptor panel:** test CCR2 and CCR6 side-by-side to establish specificity vs the known CCR2 cross-reactivity.
 3. **Antimicrobial vs chemotactic split:** MIC assays (Gram $\pm$) to anchor the core function; compare potency to chemotactic EC50.
 4. **Phylogenomics:** reconcile the *D. rotundus* beta-defensin cluster against human DEFB1/DEFB4/DEFB103 to confirm or refute DEFB1 orthology and detect lineage-specific expansion.
 5. **Structure:** AlphaFold model (AlphaFoldDB K9IFT7 exists) + electrostatics to check the CCR6-binding surface and N-terminal helix.
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Curation Leads (require curator verification)

- **Action lead:** Keep GO:0031731 only as **IEA/homology, non-core**, or down-weight/remove; add an explicit note that the sole basis is TreeGraft transfer from human DEFB1 and that CCR6 as a beta-defensin receptor is contested. **Do not** replace with generic "protein binding."
- **Candidate references to attach/verify:**
 - P 25122636

— direct human DEFB1–CCR6 interaction (supports, ortholog).
 - P 17705135

— CCR6 not a functional beta-defensin receptor (conflict/caveat).
 - P 20483750

— CCR2 as alternative receptor (caveat).
 - P 20022113

— N-terminal Leu/Ile CCR6 determinant (plausibility).

P 23411029

— primary source; salivary antimicrobial role (core function).

- **Candidate core-function terms to prioritize (better supported):** GO:0050830, GO:0050829, GO:0002227 (BP, defense/innate immunity); GO:0005576 (CC).
- **Suggested curator questions:** Is a phylogenetic-only MF receptor-binding term desired on a lineage-specific TrEMBL bat defensin? Should the annotation be qualified with a "non-core"/"inferred" note pending a direct assay?
- **Suggested experiment:** recombinant K9IFT7 CCR6-transfectant binding + chemotaxis assay (see Discriminating Tests).

Provenance Artifacts

Computed tables saved alongside the gene directory (`genes/DESR0/K9IFT7/`): -
`K9IFT7_GO_decision_table.csv` — per-term curation leads (MF/BP/CC, evidence codes, basis). -
`K9IFT7_evidence_matrix.csv` — literature evidence matrix with stance tally.

Orthology identities were computed by Needleman–Wunsch alignment of the K9IFT7 mature peptide against a UniProt-fetched human/mouse beta-defensin panel (executed in-session; DEFB1 59.1% > hBD-3 53.7% > DEFB118 52.3%). The 6-cysteine motif (inter-Cys gaps 7-5-10-7-1), net charge +4, and N-terminal Leu were computed directly from the sequence.

Limitations

TrEMBL (unreviewed) entry; **no functional data exist for any chiropteran/bat beta-defensin** (PubMed returned no hits), so every functional claim is cross-mammal homology transfer. Orthology inference used a simple global-alignment identity rather than a full phylogeny, and the human mature-peptide boundaries were not cleanly parsed for all panel members (alignments partly include propieces); nonetheless the DEFB1-closest ranking was stable across both runs, with hBD-3 (53.7%) and DEFB118 (52.3%) as near neighbors — i.e., subfamily assignment is confident but *exact* DEFB1 orthology is only moderately supported. The foundational beta-defensin–CCR6 evidence (Yang 1999,

P 10521347

used hBD-2; direct DEFB1–CCR6 evidence (Diao 2014,

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is human sperm, a non-salivary context. All computational orthology statements are inference, clearly distinguished from the primary literature.

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